

QCHPC Hybrid Application Matrix

Application Domain/ (Primary Algorithm) /Application Group	Computation Balance	Motivation / Driver	Example / Comments	Classical Counter- part	Min # of FT Qubits required (toy model/practical)	Serial/parallel/C oupling model	Associat ed Papers	Preferred hardware?	Computational Pattern
Quantum Chemistry (VQE, QPE) SOLVER	HPC-Dominant → Balanced	QC supplements HPC	QC used for molecular subproblems (e.g. Hamiltonian eigenvalues) , classical code handles orbitals, basis set construction, error mitigation, and optimization loops	Classical electronic structure methods, i.e. Classical eigensolvers (exact diagonalization, Lanczos) + electronic structure methods (CCSD, DFT)	4-8/ 50-200+	Classical: parallel; quantum: Serial. Loosely coupled (iterative hybrid loop), with emerging tightly coupled variants (QPE / adaptive VQE)	https://arxiv.org/abs/1304.3061	Superconducting (NISQ VQE) → trapped ions / FT architectures (QPE, long-term)	QOOL
Quantum Materials Science (SKQD) SOLVER	Balanced	QC-HPC balanced	Quantum Sampling of Krylov states with classical diagonalization of projected Hamiltonian	Classical Krylov subspace methods (Lanczos or Arnoldi) to find eigenvalues of large Hamiltonians	10/80+	Hybrid, strongly parallel: Krylov state sampling and measurements parallelizable; limited sequential dependence in basis construction. *Naturally <u>decomposable</u> <u>quantum</u> <u>workload</u> .	https://arxiv.org/abs/2501.09702		QASP

QCHPC Hybrid Application Matrix

Quantum Machine Learning (QML) OPTIMIZATION	Balanced → QC-Dominant	QC supplements HPC	QC performs circuit-based model training or kernel evaluation; HPC manages data loading, preprocessing, and training loop management,	Classical ML / Deep Learning / Statistical Learning	6-12/ 30-100+	Hybrid iterative tightly-coupled optimization loop	https://arxiv.org/abs/1804.00633		QOOL (dominant pattern; QML is one of the more polymorphic categories, because different subfields map differently)
AI-assisted QC Reinforcement Learning / QC Control & Optimization SOLVER	HPC-Dominant	Improve quantum system performance, calibration, control, compilation, and error mitigation	GQE (Pooja Rao) QAOA-GPT (Yuri Alexeev, Marwa)	Classical control optimization / AI-driven systems optimization	None / Early FT beneficial but not required	Tightly-coupled iterative feedback loop with real-time classical optimization	https://arxiv.org/abs/2504.16350 https://www.nature.com/articles/s41467-025-65836-3	Control-stack-integrated QPUs	QCCL
Quantum-Inspired Optimization (QAOA, VQE for Combinatorics) OPTIMIZATION/	Balanced	QC supplements HPC	QC does optimization core, HPC handles instance generation, scoring, and solution interpretation	Integer programming, Simulated annealing, Tabu search, Steepest descent, Graph neural networks	8-12/ 30-300+	Serial classical optimization with parallelizable quantum sampling/evaluation kernels	https://arxiv.org/abs/1304.3061		QOOL (dominant) with some overlap into: QSMC when sampling dominates runtime.
Quantum linear systems algorithm (HHL, VQLS) SOLVER	HPC-Dominant → Balanced	QC supplements HPC	QC used to solve linear system of equations	Classical sparse linear solvers / iterative	2-5 / 50-100+	Quantum accelerator-style linear algebra kernel with classical	https://arxiv.org/abs/0811.3171		QLAO Quantum linear algebra accelerator kernel

QCHPC Hybrid Application Matrix

				Krylov-sub space methods (CG, GMRES, multigrid, sparse direct methods)		preprocessing/postprocessing	https://arxiv.org/abs/1909.05820		
Hamiltonian Simulation (Lattice Models, Spin Chains) PHYSICS SIMULATION	Balanced	QC supplements HPC	QC runs core quantum time evolution or state prep; HPC manages grid construction, observable evaluation, and visualization	Classical many-body simulation / tensor network methods / exact diagonalization	6-10/50-200+	Contiguous quantum time-evolution kernel with classical pre/post-processing	https://www.science.org/doi/10.1126/science.273.5278.1073 https://link.springer.com/article/10.1007/BF02650179		QTES — Quantum time-evolution and simulation kernel
Quantum Circuit Compilation and Verification QUANTUM INFO PROCESSING	HPC-Dominant	HPC supplements QC	HPC used to simulate, verify, and optimize quantum circuits before execution; QC is target system Alphatensor for quantum circuit optimization (important for genesis)	Classical compiler/runtime optimization infrastructure	N/A,N/A	Predominantly classical parallel workflow with iterative quantum hardware validation	https://arxiv.org/abs/1809.02573 https://arxiv.org/abs/1904.06319		QCVW

QCHPC Hybrid Application Matrix

Quantum Finance (Portfolio Optimization, Option Pricing) OPTIMIZATION	Balanced	QC supplements HPC	Quantum component handles the core combinatorial problem or simulation, HPC handles data input and analysis	Large-scale stochastic financial simulation and optimization	8-12/100-300				QASP, with secondary overlap into QOOL/QSMC
Quantum Control and Pulse Optimization OPTIMIZATION	HPC-Dominant	HPC supplements QC	Classical optimization algorithms used to shape pulses, feedback loop leverages HPC	Classical optimal control and pulse shaping	N/A//N/A				QCCL
Quantum-assisted Monte Carlo with Quantum Sampling Subroutine SOLVER	HPC-Dominant	QC supplements HPC	QC does rare-event or high-dimensional sampling, HPC does standard sampling and aggregation	Classical stochastic sampling and statistical estimation	6-10/50-100	Parallel Monte Carlo workflow with batched QC sampling			QSMC

QCHPC Hybrid Application Matrix

Hybrid Training of Quantum Neural Networks OPTIMIZATION	Balanced → QC-Dominant	QC supplements HPC	QC computes gradients or state overlaps; HPC handles batching, optimizer updates	Classical neural network training and deep learning optimization	6-12/40-100	Tightly-coupled hybrid training loop. Hybrid optimization workflow with repeated quantum circuit evaluation and classical parameter updates			QOOL
Quantum-Assisted Data Compression or Encoding QUANTUM INFO PROCESSING	QC-Dominant	QC supplements HPC	Quantum encoding of large datasets (e.g., QRAM-like structures); classical computation provides source data	Classical data compression and encoding; feature extraction, dimensionality reduction	10-20/100-500	Parallel preprocessing with quantum encoding kernels			QDSP
Quantum Annealing + Classical Pre/Post-processing OPTIMIZATION	Balanced	QC supplements HPC	Classical code used to embed, decode, and interpret; quantum system used for core annealing		8-16/1000+				QAHM
Quantum Volume / Benchmark Characterization QUANTUM INFO PROCESSING	HPC-Dominant	HPC supplements QC	HPC handles simulation of circuits, statistical aggregation; QC only runs		4-8/20-50+				

QCHPC Hybrid Application Matrix

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